SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA16	**Course Title:**	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	**Assessment:**	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	**Total Marks:**	100 (5 Questions × 20 Marks)
CO1 Statement:	*Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2*		
Student Name:	S. ROOBINI	**Reg. No.:**	192421114
Section / Batch:	IT - 2024	**Date:**	15/07/2026

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
		GRAND TOTAL:	100 Marks

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse for City Hospital consists of the following four dimensions: patient, doctor, department, and time, and two measures visit_count and avg_charge. At the lowest conceptual level, avg_charge stores the actual consultation fee charged for a patient visit. At higher conceptual levels, it stores the average fee.

(a) Draw a snowflake schema diagram for the data warehouse.

(b) Starting with the base cuboid [patient, doctor, department, time], what specific OLAP operations should be performed to list the average consultation fee department-wise for each year?

(c) If each dimension has four levels (including all), how many cuboids will this cube contain (including base and apex cuboids)?

2. A smart city project is collecting data from multiple sources, including:

- Traffic sensors for vehicle counts and congestion levels.
- IoT devices for monitoring air quality, noise levels, and energy consumption.
- Public transportation systems for ridership data and punctuality.
- Citizen feedback from a mobile app.

Questions:

Fact and Dimension Design:

- Design a star schema to analyze traffic congestion patterns, energy consumption trends, and public transportation efficiency.
- What measures and dimensions would you include? Justify your design.

3. Suppose that a banking data warehouse contains two fact tables: Transaction_Fact and Loan_Fact, sharing dimensions customer, branch, time, and account_type.

- Transaction_Fact → transaction_count, amount
- Loan_Fact → loan_amount, interest

(a) Draw a galaxy schema diagram.
(b) Identify OLAP operations to obtain branch-wise average transaction amount and loan amount yearly.
(c) If each dimension has four levels, calculate total cuboids.

4. A manufacturing company monitors product quality and production efficiency. The data warehouse contains:

- Dimensions:
  - Product: Category, type, and batch.
  - Factory: Location, department, and line number.
  - Time: Year, quarter, month, and week.
  - Supplier: Region, rating, and contract type.
- Measures:
  - Defect rate.
  - Production units.
  - Supplier reliability percentage.

Questions:

- Roll-up: Summarize defect rates by product category for the current year across all factories.
- Drill-down: Focus on a specific factory to analyze defect rates by department and product type.
- Slice: Isolate production unit data for suppliers with a rating of 4+ across all regions.
- Dice: Analyze defect rates and supplier reliability for selected product categories and factory locations.

5. A retail chain manages inventory and sales data across multiple stores. The data warehouse contains:

- Dimensions:
  - Product: Category, subcategory, and SKU.
  - Store: Region, city, and store ID.
  - Time: Year, quarter, month, and day.
  - Customer: Membership tier, age group, and gender.
- Measures:
  - Inventory turnover.
  - Total sales revenue.
  - Units sold.

Questions:

1. Roll-up: Summarize inventory turnover by product category across all stores for the last quarter.
2. Drill-down: Focus on sales data for electronics in a specific region, analysing by subcategory and SKU.
3. Slice: Isolate sales revenue data for male customers in the Bronze membership tier across all stores.
4. Dice: Analyze sales revenue and units sold for selected cities and product categories during specific time periods.

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Marks Evaluation:

Question No.	Understanding of Concepts (25%)	Explanation (25%)	Application (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total Marks
Q1 (20 Marks)							
Q2 (20 Marks)							
Q3 (20 Marks)							
Q4 (20 Marks)							
Q5 (20 Marks							
						Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ______________	Signature: ______________	Signature: ______________
Date: __________________	Date: __________________	Date: __________________

CSA1610 - DATAWAREHOUSING AND DATA MINING

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ASSESSMENT TOOL - 1

SHORT ANSWERS.

1. a) Snowflake Schema.

Dimension table: Fact Table: Hospital_Visit_Fact.

- Patient → Patient_ID → City → state → country
- Doctor → Doctor ID → Specialization → Qualification
- Dept → Dept ID → Dept.name → Hospital.

b) OLAP operations:

1. Roll-up time from day → year
2. Roll-up dept if needed to dept level
3. Aggregate avg-charge using avg
4. Display Dept x Year.

c) No. of. cuboids:

Formula: $(L1 \times L2 \times L3 \times L4)$

4 level: $(4 \times 4 \times 4 \times 4) = 256$

∴ 256 cuboids.

2. a) Star Schema : Fact table

Traffic-fact

- Vehicle-count
- Congestion-level
- Energy-Consumption
- Ridership
- Punctuality

Dimension tables

- Time
- Location
- Sensor
- Transport_mode
- Citizen.

6. Measures & Dimensions

Measures	Dimensions
◦ vehicle count	• Time
• Congestion level	• Location
• Energy consumption	◦ Sensor type
◦ Ridership	◦ Transport mode
• Punctuality	• Citizen
◦ Feedback	

3. a) Galaxy Schema

Fact tables :

Transaction - Fact

- transaction - Count
- amount

Loan - Fact

- loan - amount
- interest

Shared dimensions :

- Customer
- Branch
- Time
- Account - type.

b) OLAP Operations :

1. Roll-up time → year
2. Roll-up branch → Branch level
3. Calculate :
   - Avg Transac Amt .

Total cuboids.

4 dimensions x 4 levels.

$4^4 = 256$.

Ans: 256 cuboids.

A. Manufacturing Company data warehouse.

1. Roll-up.

Roll-up is OLAP operation that moves from detailed data to summarized data by climbing up a dimension heirarchy.

Operation:

- Product: Batch → Type → category.
- Time: Filter current Year
- Factory: Roll up to All factories.

2. Drill-Down:

Drill-Down moves from summarized info to more detailed info.

Operation:

- Select one factory
- Expand data:
  - Factory → Dept
  - Product → Type.

3. Slice :

A Slice select a single value or cond from one dimension while keeping the remaining dimension available for analysis.

operations:

- Supplier Rating ≥ 4
- Display Production units.
- Include all regions

4. Dice.

A Dice operation selects a subset of data using multiple conditions across multiple dimension.

Eg selection :

- Product categories:
  - Electronics
  - Furniture
- Factory location :
  - Chennai
  - Bangalore.

5. Retail chain Data Warehouse.

1. Roll-up :

Roll-up summarizes detailed data into

higher - level info.

operation:

- Product SKU → Sub category → Category
- Store : All stores
- Time : Last Quater.

2. Drill - down.

Drill - down provides more detailed info.

Operation:

- Select electronics
- Select South Regions
- Expand:

Category → Subcategory → SKU.

3. Slice.

A Slice selects a specific value from one or more dimensions.

operation:

- Gender : Male
- Membership Tier : Bronze.

Dice :

4. A Dice operation filters data using multiple condition across several dimension

Eg Selection :

- cities :
  - Chennai
  - Madurai
- Product Categories :
  - Electronics
  - Grocery
- Time .
  - Jan - Mar 2026 .

Measures :

- Total Sales Revenue .
- Unit Sold .